

Basic course

System overview

KNX Association

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Preface

This training document is created for the Basic course. Sections with a heading marked [E] contain important information that is also required for the NextGen course. This includes any sub-sections within them. If a heading does not have [E], it means the content is only for reference and is not mandatory for the NextGen course.

The documentation focuses on the most used configuration mode in KNX, i.e., S-Mode (System Mode). Configuration modes that used to be supported in the past like E-Mode (Easy Mode) are not covered further.

1 Definition [E]

The following terms are used as synonyms in KNX literature:

Terms used in the KNX training documentation and in ETS	Alternative term
Individual address	Physical address
Group object	Communication object

2 Minimal structure of a KNX TP installation [E]

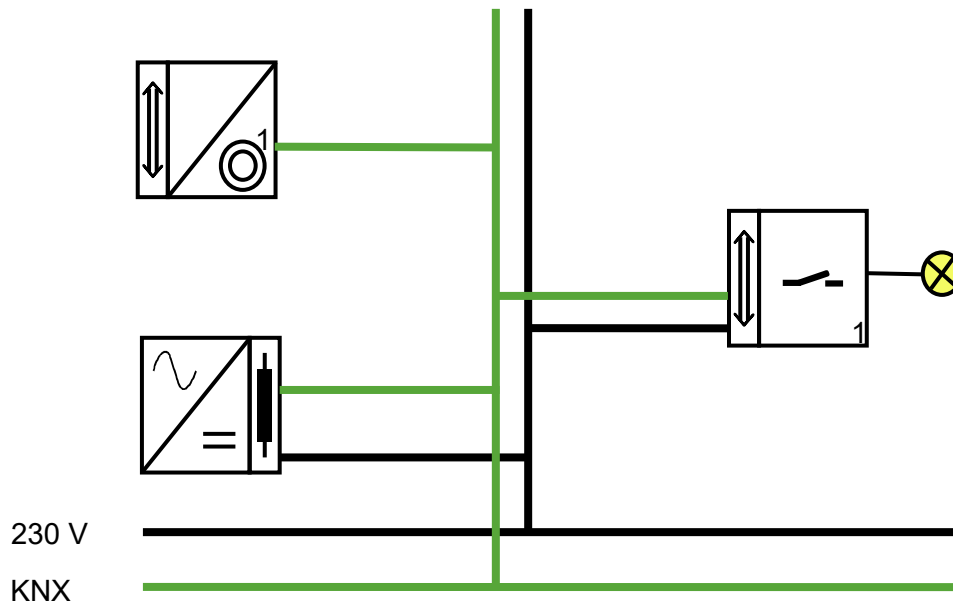


Figure 1: Minimal structure of a KNX TP installation

A KNX TP installation requires minimal the following essential components:	
1 KNX power supply unit (DC 30 V) with choke	The choke is usually built into the power supply unit.
1 sensor	- A single push button with one rocker or two buttons is represented in the figure above. - Sensors are powered by the KNX power supply unit.
1 actuator	A single switch actuator is represented in the figure above.
Bus cable	- Only two wires of the bus cable are required to exchange data (telegrams) and in most cases to power the bus devices. - In the figure above, the bus cable is represented as a green line. - It connects sensors, actuators and the KNX power supply unit. - It can also provide extra power to bus devices using the spare wire pair.

3 Addressing

In KNX there are two types of addressing, i.e., individual addressing and group addressing.

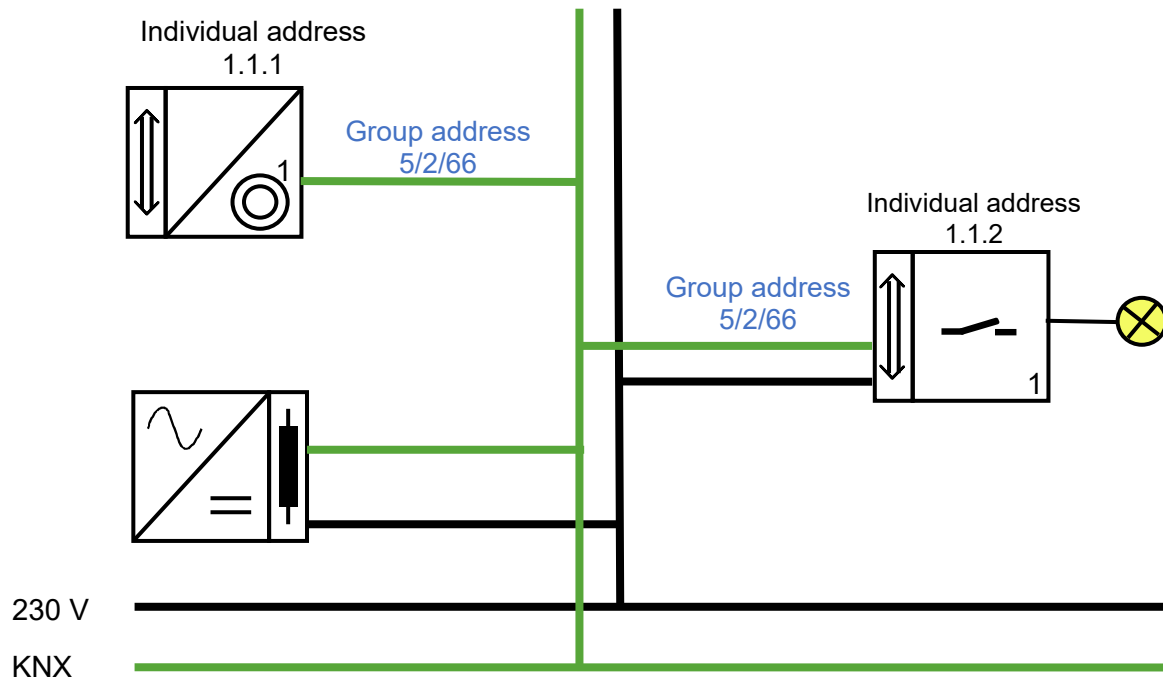


Figure 2: Addressing

3.1 Individual address [E]

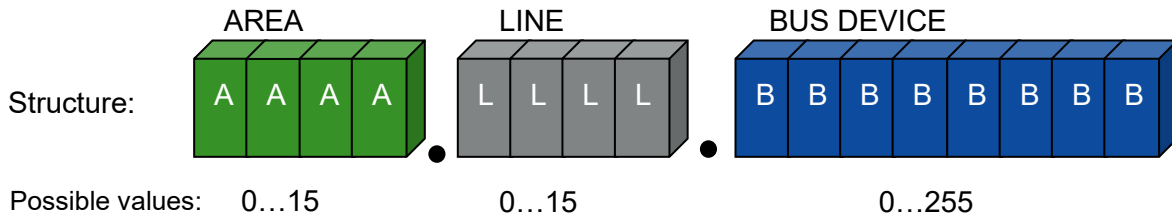


Figure 3: Structure of the individual address

Each device in a KNX system needs a unique individual address to ensure proper communication and avoid conflicts. The individual address allows ETS (Engineering Tool Software) to send programming telegrams, new application or parameter data to the bus device.

Structure of the individual address:

- The individual address is a 16-bit value with a fixed format.
- It is represented in decimal format in ETS and KNX documentation, using two dots as separators (A.L.B).

Assigning an individual address:

- Manual method:
 - The bus device must be set to programming mode before assigning an individual address.
 - The programming mode must be activated on the bus device via pressing the programming button or using the method provided by the manufacturer.
 - When in programming mode, the LED on the device lights up, the individual address will be downloaded to the device. The LED will turn off afterwards.
- Automatic method:
 - Enter the individual serial numbers of the bus devices in the ETS project.
 - When programming the individual address, ETS now asks for the programming method (in this case, select "Use Serial Number").
 - ETS can now download the individual address to the device (similar to DHCP).
- Storing the individual address in the device:
 - Once assigned, the individual address is saved permanently in the device.
 - ETS then uses this address to transfer all necessary data, including:
 - Application and configuration settings
 - Parameter values
 - Group address assignments

After installation and setup, all normal communication (e.g., turning lights on/off) happens via group addresses, not individual addresses.

3.2 Group address [E]

3.2.1 Introduction to group addresses [E]

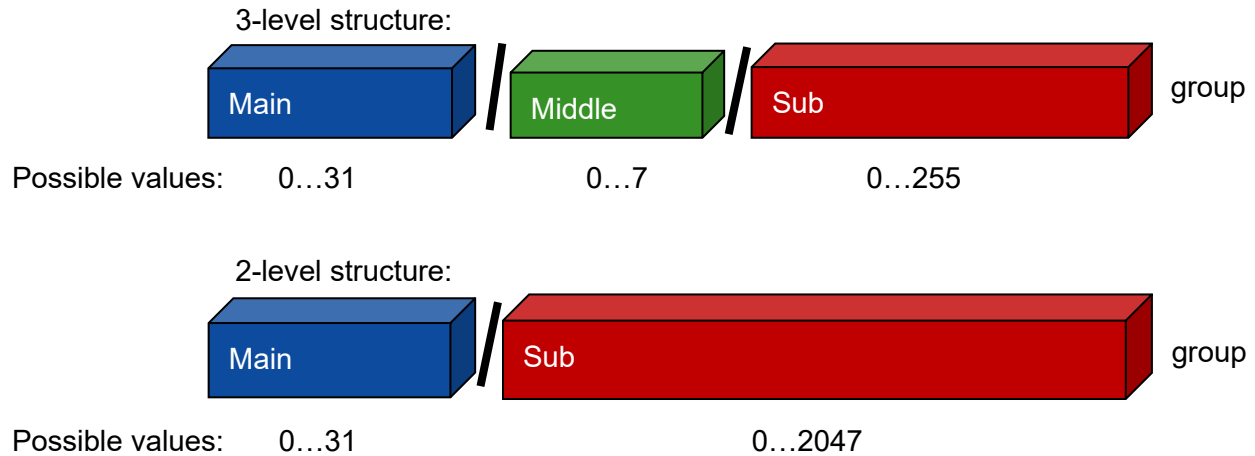


Figure 4: Structure of group addresses

In a KNX installation, normal communication between devices happens through group addresses. These addresses allow multiple devices, such as sensors and actuators, to interact and perform functions like turning lights on/off or adjusting heating.

The ETS user assigns a group address for each function in the installation. The structure of the group addresses can be freely defined based on the project requirements.

A KNX system supports up to 65,535 group addresses that can be created and linked in ETS.

By properly creating group addresses, devices in a KNX system can efficiently communicate and execute functions.

Important rules for group addresses:

- A group address range starts with 0/0/1.
- The group address "0/0/0" is reserved for broadcast telegrams.
 - It is used for assigning individual addresses.
 - All bus devices acknowledge receipt of a broadcast telegram.

Group address representation in ETS:

In ETS, group addresses can be organized using either:

- **3-Level structure:** Main Group/Middle group/Subgroup (default setting)
- **2-Level structure:** Main Group/Subgroup
 - The level structure does not affect device communication; it is only used for better organization and clarity in ETS projects.
 - The 3-level structure is set as default but can be changed in the project details of ETS project itself.

Example of a 3-level group address structure:

Group address	Function
5/2/66	Room 523, switch light 1.
5/2/67	Room 523, switch light 2.
5/2/68	Room 523, switch all the lights together.
5/2/69	Switch lighting on the 2nd floor in building number 5.

In KNX, the meaning of each group address level can be **freely** defined by the ETS user based on the project requirements. The structure behind the above example is:

- Main group = building number (e.g., building no. 5) and
- Middle group = floor number (2nd floor).

However, a commonly used structure follows this approach:

Level	Definition	Example
Main group	Floor	1/ (Ground floor)
Middle group	Functional domain (e.g., lighting, heating)	1/3/ (Blinds on ground floor)
Subgroup	Specific function or device group	1/3/66 (Move blinds up/down in room A on the ground floor)

Common middle group assignments:

- 1 = Lighting
- 2 = Heating
- 3 = Shading (blinds, shutters, curtains, etc.)
- 4 = Ventilation
- 5 = Security (alarms, access control, etc.)

This structured approach helps keep group addresses organized and easy to manage in ETS, improving efficiency in large installations.

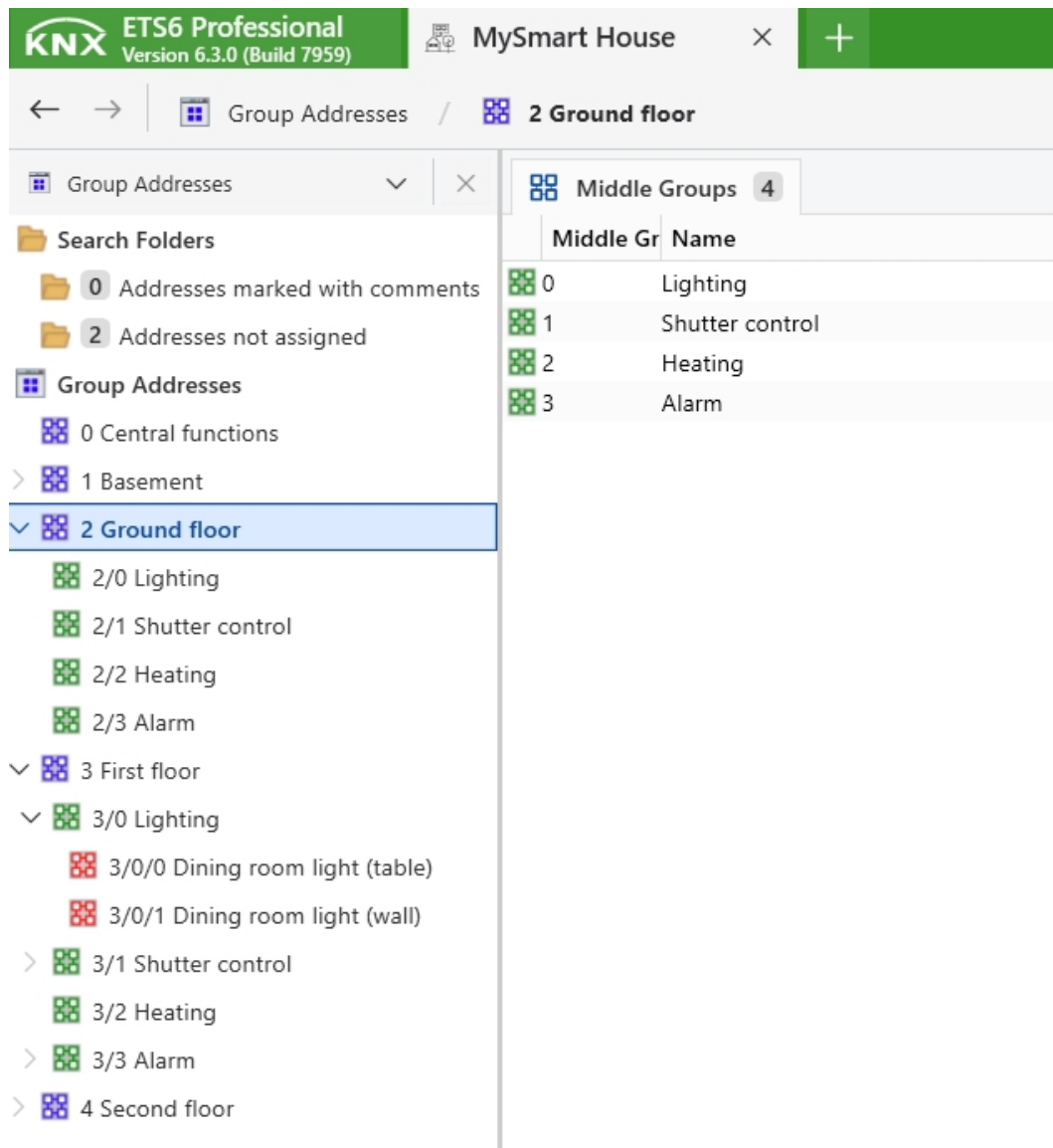


Figure 5: Example: structure of group addresses in ETS

To ensure consistency and easier management across multiple projects, it is recommended to define a company-wide default group address structure and use it in all KNX projects. This helps ETS users and technicians quickly understand and navigate different projects.

The group addresses are assigned to the group objects of the respective bus devices, with the help of ETS.

Summary: Individual vs. group addressing in KNX:

- The **individual address** is mainly used during **commissioning and diagnostics** in **ETS**, allowing ETS users to configure and program specific devices.
- **Group addressing** is the primary method of communication during **normal operation**, as it enables interaction between multiple devices.
- Once commissioning is complete, the individual address becomes less important, while group addresses handle all functional communication.

Comparison: Individual address vs. group address:

Address type	Application	Analogy (letter/post example)
Individual address	Destination address in ETS “programming telegrams” for transmitting new application and parameter data to a single bus device.	Addressed to a specific person (e.g., "John Smith, Samplestreet 12, US-12345 Tinseltown").
Group address	Destination address used during “normal” operation for telegrams like “Lighting room 424 ON/OFF”.	Works like bulk mail, sent to multiple recipients (e.g., "To all households with a photovoltaic installation").

Important notes:

- Actuators can listen to and react to multiple group addresses for each group object.
- Sensors, however, can only send one telegram with a specific group address per group object.
- For centralized control, a single group address is used:
 - It is linked once to a sensor (e.g., a central switch).
 - It is linked multiple times to all connected actuator group objects (e.g., all lights in a building turning on/off together).

3.3 Configuration steps for KNX devices [E]

After mounting the devices, a KNX installation is not operational until the sensors and actuators have been downloaded with the application program using the ETS program. The ETS user must first complete the following configuration steps in ETS:

- **Assign individual addresses**
 - Each sensor and actuator must be given a unique individual address for identification within the KNX installation.
- **Select application program**
 - Choose the appropriate application program for each bus device to define its functionality.
- **Set device parameters**
 - Configure specific parameters for each device based on project requirements (e.g., switch delay times, dimming levels, temperature thresholds).
- **Assign group addresses**
 - Link sensors and actuators via group addresses to enable communication and perform the desired functions (e.g., turning lights on/off, adjusting blinds).

3.4 How the KNX installation functions after commissioning [E]

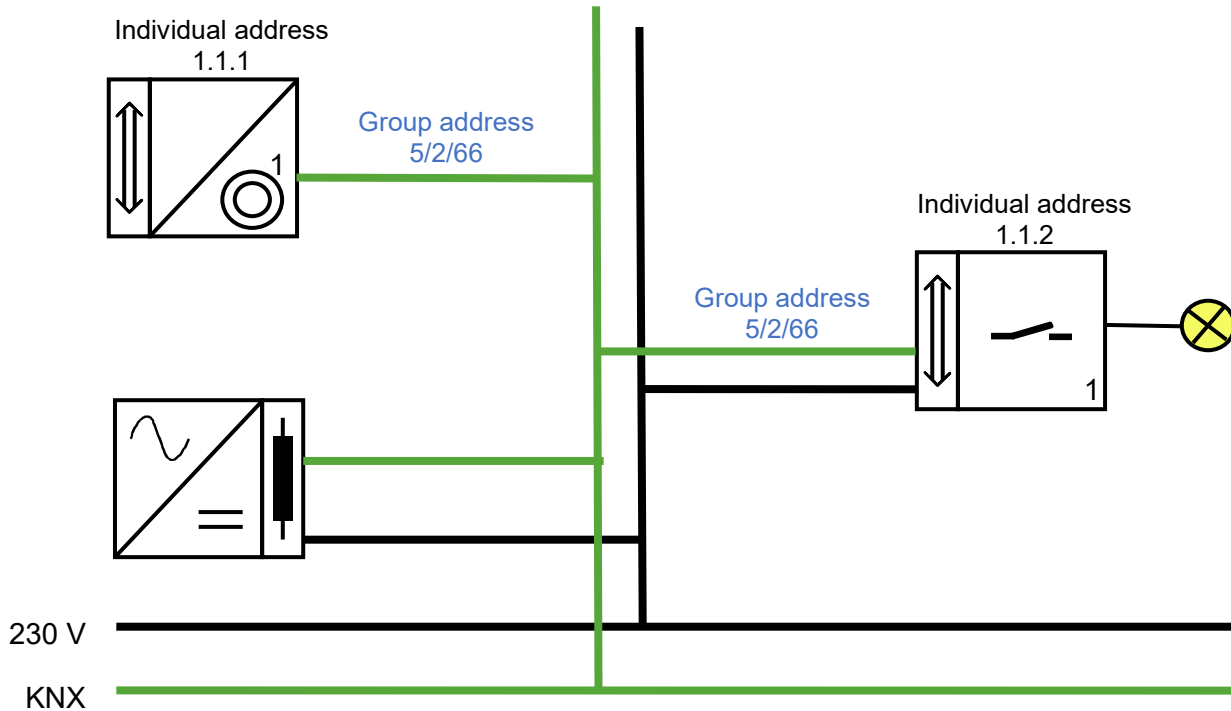


Figure 6: Function after commissioning stage

Once the KNX installation has been fully configured, the system operates as follows:

1. Sending a command

- When the upper part of the rocker switch on the single push button (1.1.1) is pressed, it:
 - Sends a telegram containing:
 - Group address: 5/2/66
 - Control value: "1" (Switch ON)

2. Receiving and processing the telegram

- The telegram is broadcasted to all connected KNX bus devices.
- Any device that has 5/2/66 in its group address table will:
 - Acknowledge the telegram (confirming correct or incorrect reception).
 - Read the control value and execute the command accordingly.

3. Activating the actuator

- In this example, the switch actuator (1.1.2) also has the group address 5/2/66 assigned to it.
- It closes its output relay, switching the connected load (e.g., a light) ON.

4. Turning the device off

- When the lower part of the rocker switch is pressed:
 - A new telegram is sent with:
 - Group address: 5/2/66
 - Control value: "0" (Switch OFF)
- The switch actuator (1.1.2) then opens its output relay, turning the light OFF.

4 Group objects in KNX devices [E]

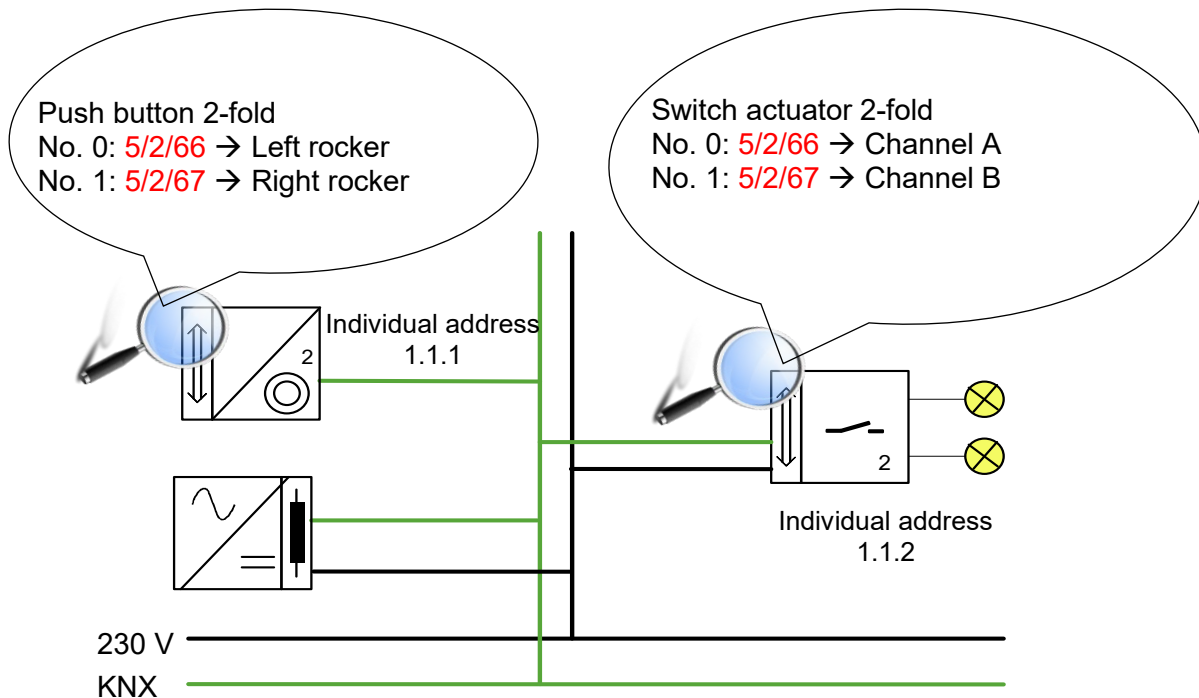


Figure 7: More detailed description of bus devices with group objects

In the earlier example, a group address was assigned directly to a single push button and a single-channel actuator. However, in real-world installations, things are often more complex.

Understanding group objects:

- Many bus devices (such as push buttons and actuators) have multiple channels that can communicate. For example:
 - A push button can have multiple rockers, each able to control a different function.
 - A switch actuator can have multiple switching outputs controlling different loads.
- To handle this, KNX uses "group objects"; these represent specific functions within a device.

What are group objects?

- Group objects are memory locations inside a KNX device, used to store and process data.
- They act as the interface between the device and the KNX bus, determining how it reacts to incoming signals.
- The size of a group object – defined by the manufacturer - **can vary from 1 bit to 14 bytes**, depending on the function. Some examples are provided in the following table:

Size	Purpose
1 bit	For simple ON/OFF switching (e.g., turning lights ON/OFF).
1 byte	For dimming control or HVAC settings (e.g., brightness level, percentage value).
2 bytes	For sending temperature values or other numerical data.
4 bytes	For numeric high-resolution data (e.g., metering values).
14 bytes	For bulk data like text messages (e.g., sending display messages).

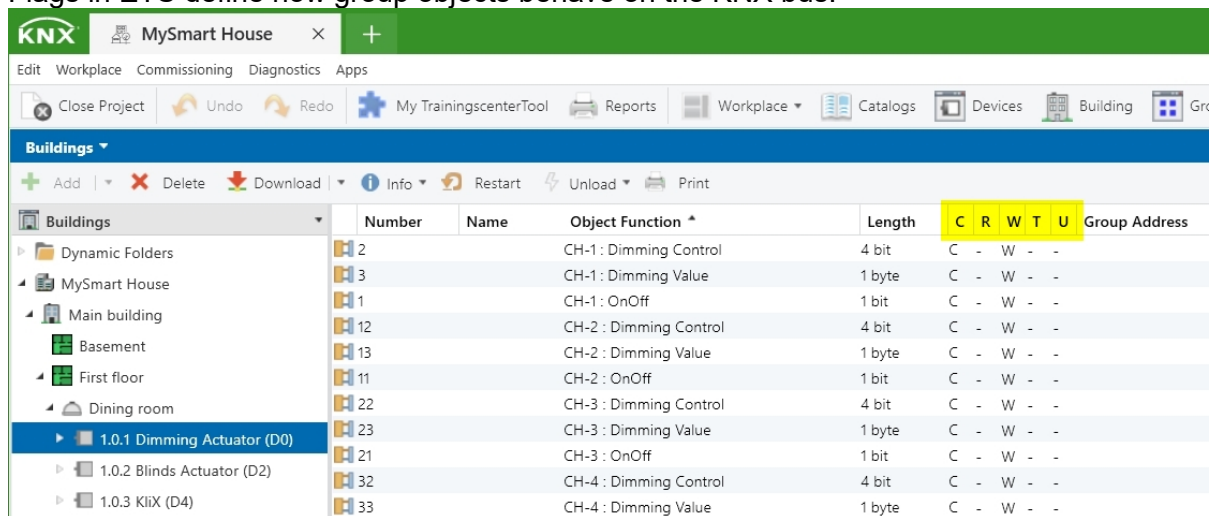
Linking group objects via group addresses:

ETS ensures that only group objects of the same size can be linked together via group addresses. For example, a 1-bit push button rocker switch can only be linked to a 1-bit switch actuator output. Figure 7 shows the relation using a push button 2-fold and a switch actuator 2-fold as an example.

4.1 Properties of group objects

4.1.1 Flags in ETS

Flags in ETS define how group objects behave on the KNX bus.

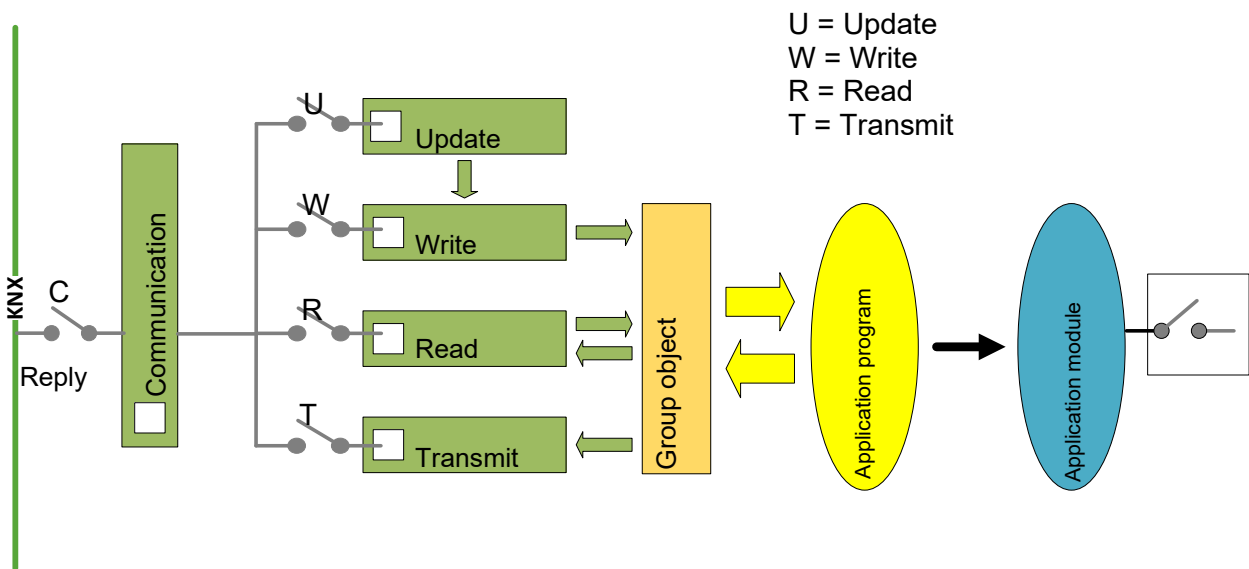


Buildings	Number	Name	Object Function ^	Length	C	R	W	T	U	Group Address
Dynamic Folders	2	CH-1 : Dimming Control	4 bit	C	-	W	-	-	-	
MySmart House	3	CH-1 : Dimming Value	1 byte	C	-	W	-	-	-	
Main building	1	CH-1 : OnOff	1 bit	C	-	W	-	-	-	
Basement	12	CH-2 : Dimming Control	4 bit	C	-	W	-	-	-	
First floor	13	CH-2 : Dimming Value	1 byte	C	-	W	-	-	-	
Dining room	11	CH-2 : OnOff	1 bit	C	-	W	-	-	-	
1.0.1 Dimming Actuator (D0)	22	CH-3 : Dimming Control	4 bit	C	-	W	-	-	-	
1.0.2 Blinds Actuator (D2)	23	CH-3 : Dimming Value	1 byte	C	-	W	-	-	-	
1.0.3 Klix (D4)	21	CH-3 : OnOff	1 bit	C	-	W	-	-	-	
	32	CH-4 : Dimming Control	4 bit	C	-	W	-	-	-	
	33	CH-4 : Dimming Value	1 byte	C	-	W	-	-	-	

Figure 8: Flags in ETS

4.2 The 6 basic flags

Flag	Function	Description
C - Communication	Enables telegram transmission	If set, the object can send and receive telegrams. This is the master switch.
R - Read	Allows reading values	If enabled, other devices can request the object's current value. It will send a Response telegram.
W - Write	Allows writing values	If enabled, the object can be updated by other devices.
T - Transmit	Sends values on the bus when they change	If the value changes, it is immediately sent to the bus.
U - Update	Interprets reply telegrams as Write telegrams	Forces objects to update their values when a reply is received.
I - Read on Init	Sends a Read Value request after voltage recovery	Ensures re-initialization of values after a power failure.



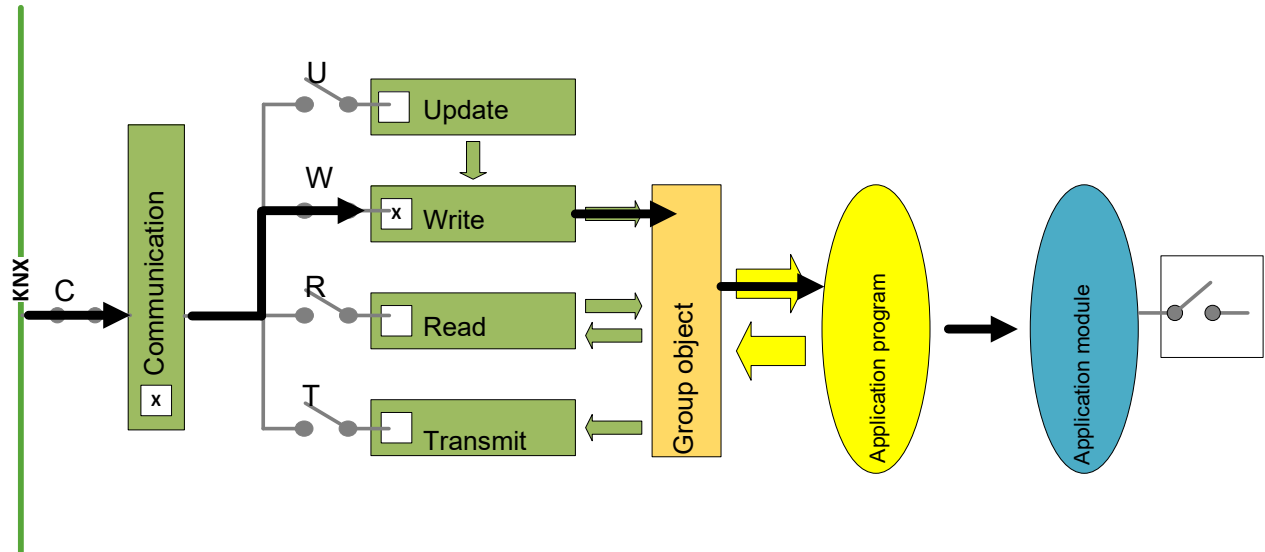
Note: Even if the Communication flag is deactivated, telegrams are still acknowledged.

4.3 Additional notes on flag behavior

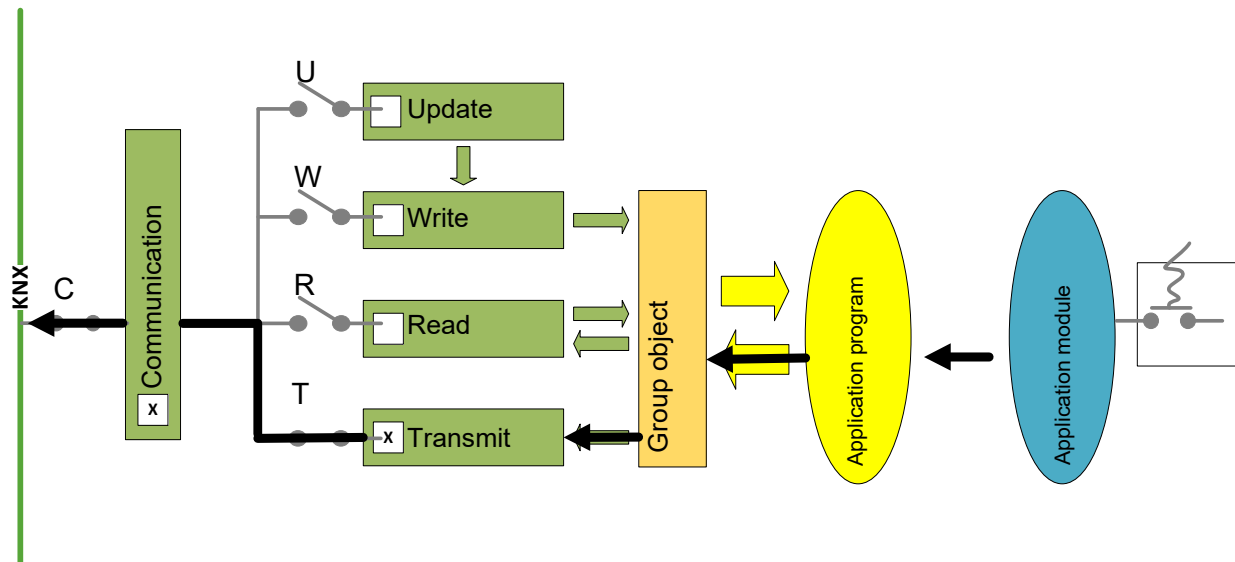
- Not all objects need all flags:
 - Example: A sensor **may** have C, T, and W flags, but not R (as it does not need to be read).
 - Example: An actuator may have C, R, and W flags, but not T (as it reacts to changes but does not transmit them).
- Flags help optimize bus communication:
 - Unnecessary flag settings can increase bus traffic, reducing system efficiency.

Below you will find the 4 most important basic settings, depending on the object type:

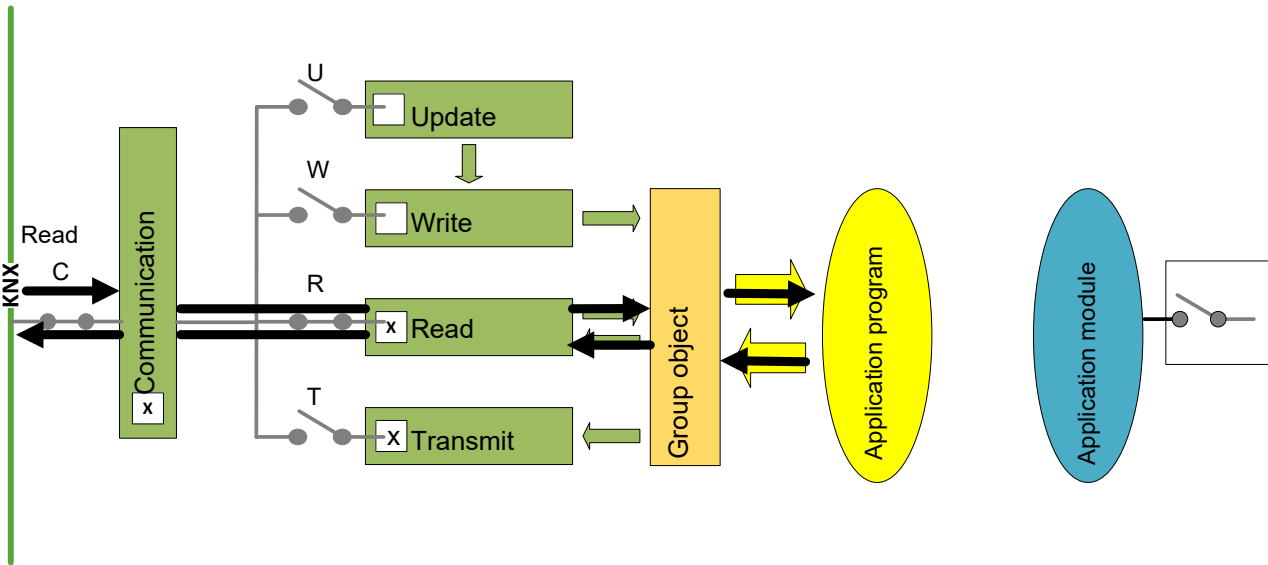
4.4 Minimum settings of the flags for actuator objects



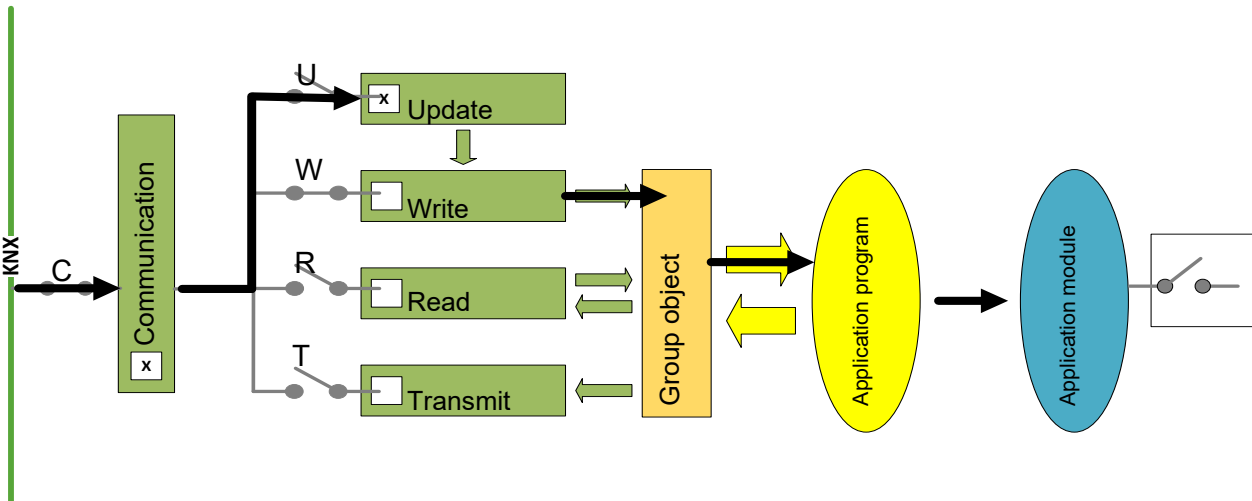
4.5 Minimum settings of the flags for sensor objects



4.6 Typical settings of the flags for status objects



4.7 The Update flag



Update flag - behavior:

- If the Update flag is set, a reply telegram will be interpreted as a Write telegram.
- System 1 devices: The Update flag is always active; it cannot be deactivated.
- System 2, System 7, and System B devices: The Update flag can be deactivated if necessary.
- **Purpose:** Ensures that reply telegrams automatically update the object's value when received.

Read at Init flag (System B):

- Available for certain system profiles (System B devices).
- Function: If set, the group object automatically sends a Read Value telegram after voltage recovery.
- **Purpose:** Ensures that the object initializes its value correctly after a power failure.

Important notes on KNX group object flags:

- Flags set by the manufacturer should only be changed if necessary.
- Only modify flags if specific application requirements demand it.

For a detailed explanation of the flags, refer to the "Flags" chapter in the KNX Advanced course.

A group communication in detail:**Sending a command from a push button:**

- When the upper part of the left rocker on a 2-fold push button is pressed, the following happens:
 - The push button writes a control value of "1" to its internal group object (Object number 0).
 - The system software ("firmware") of the push button ensures that a telegram is sent onto the KNX bus.
 - The group object has the following flag settings (C and T flags are set).
 - The telegram contains the following information:
 - Group address: 5/2/66
 - Command type: "Write value"
 - Value: 1 (meaning "Switch ON")

Receiving and processing the telegram:

- All KNX bus devices in the system that have group address 5/2/66 assigned will process this telegram:
 - These devices detect the telegram on the KNX bus.
 - Each device checks if it has group address 5/2/66 in its address table.
 - If the group address matches, the device updates its own group object with the value 1.
 - In this example, the switch actuator will take over the control value "1" in its group object (Object number 0).

Executing the command on the actuator:

- The application software of the actuator detects the new value in its group object and executes the required action:
 - The group object requires the following flag settings (C and W flags are set).
 - It notices that the group object value has changed from 0 to 1.
 - Based on its function, it closes its output relay, turning the connected load (e.g., a light) ON.

Summary of the process:

Step	Action	Example
1. User input	Push button is pressed.	The upper part of the rocker is pressed.
2. Send telegram	Push button writes "1" to its group object and sends a group telegram containing the useful data "1".	5/2/66 - Write Value - 1
3. Receive telegram	Devices linked to 5/2/66 update their own group object.	Switch actuator stores 1 in its group object memory.
4. Execute action	The actuator detects a value change and switches ON.	Light turns ON.

5 Useful data of a TP telegram [E]

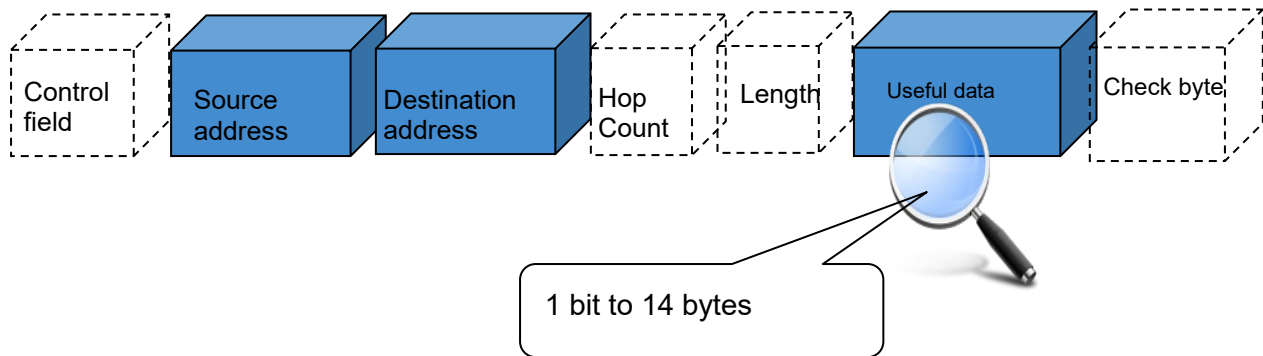


Figure 9: Useful data of a TP telegram

A TP telegram in KNX contains useful data, which varies in size depending on the **Datapoint type** (DPT) used. The size can range from 1 bit to 14 bytes as explained earlier.

6 KNX Secure

A KNX installation is by default secure as the data sent/received on the bus cannot be interpreted just like that from the telegram traffic. This, however, does not mean that a KNX installation doesn't have to follow a set of security rules to prevent unauthorized access to the KNX installation.

6.1 Security measures

6.1.1 First level

The first level security measures can be taken by the KNX installer during the KNX installation:

- The installer needs to ensure that KNX devices cannot be unmounted both inside and outside of the building. If mounted outside, these devices (e.g. weather station, movement detector etc.) shall be mounted at sufficient height. Moreover, the TP cable may not be visible in- or outside the building.
- Unauthorized people shall have no access to the distribution board.
- Any touch panels in public areas shall be password protected, and certain functions should only be accessible in “guest” mode.
- A BCU key shall be set in ETS projects. This will ensure that no unwanted parameter changes are downloaded to KNX bus devices. If that happens, ETS will prompt the user to enter the correct BCU key (not known to hackers).
- **Note:** Manufacturers can by means of a BCU Key realize a memory read/write protection of their application programs. The BCU Key is an ETS project property and will, if set, during any download be applied to any device of the project supporting this system feature. The device types supporting BCU Key based authorization are System 2, System 7, and System B.

Beware: if the set BCU key is lost, the devices with a set BCU key will no longer allow to redownload them.

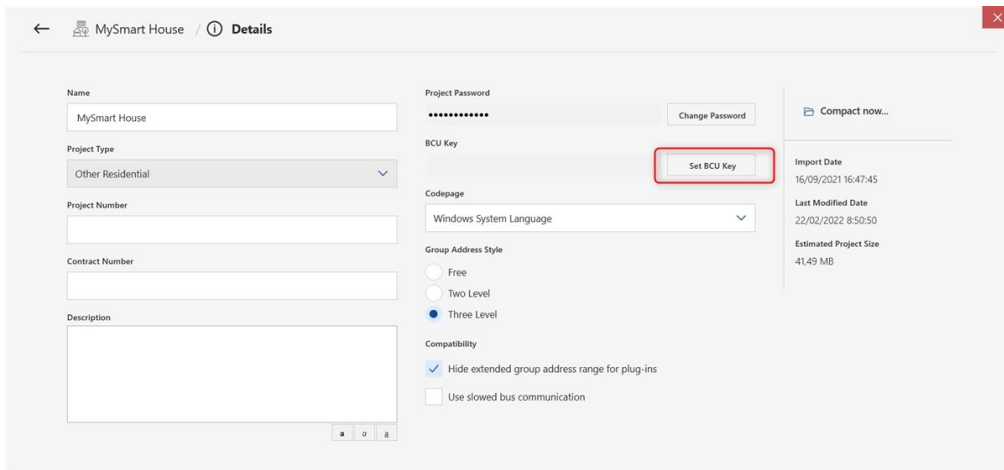


Figure 10: Set BCU Key in ETS project

- The KNX line couplers or IP routers must be configured in such a way to prevent unwanted telegram traffic in parts of the installation outside this line.
- ...

6.1.2 Second level

The second level requires a certain knowledge of IT from the KNX installer and/or the KNX building owner. If, e.g., the installer wants to access the KNX installation remotely to further optimize the ETS project, then he/she needs **remote access**. The same goes for the owner who wants to control his building remotely (e.g. lowering the shutters by means of a smartphone, monitoring consumption data on tablet etc.). Such installations require a KNXnet/IP Router or a KNX device that supports secure remote access.

- Opening port 3671 (IP port used for KNXnet/IP communication) has historically been used to provide remote access to a KNX system. As this port is standard it can be misused more easily by unauthorized people. Opening this port on your router is if like leaving the front door of your house wide open for unwanted intruders. **Therefore, KNX strictly advises against opening port 3671 for KNXnet/IP.**
- If remote access is required, then the KNX installer needs to ensure that access is only possible from the Internet (IP) side to the KNX installation via a VPN (Virtual Private Network) connection. Certain routers natively support a VPN connection. If your router does not natively support VPN, consider buying a router that supports VPN or integrate a KNX IP device that natively supports secure remote access.
- Telegrams sent from the bus to the IP network (or vice versa), travel as multicast telegrams thereby using the default multicast IP address 224.0.23.12. This is a for KNX Association officially registered IP multicast address. To limit unauthorized access, it is possible to use a different multicast address in ETS.
- The building owner shall ensure that the default wireless network password has been changed to a stronger WPA2¹ password.
- ...

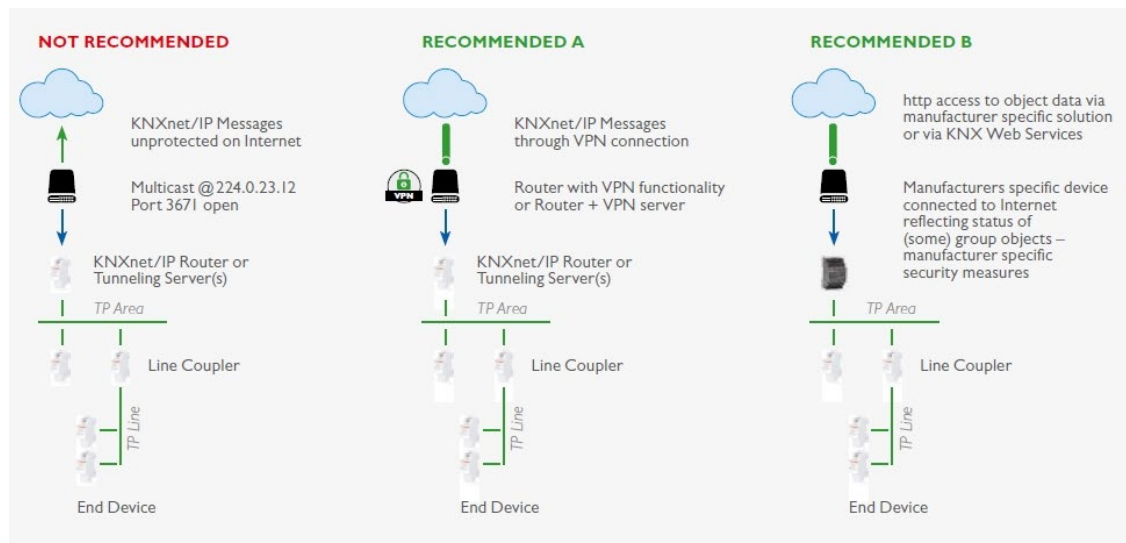


Figure 11: Access to the KNX network via Internet

¹ WPA stands for Wi-Fi Protected Access and is a security standard that protects wireless networks. WPA2 is the second generation of the Wi-Fi Protected Access security standard and so is more secure than its predecessor, WPA.

6.1.3 Third level

Next to some of the above-mentioned measures, KNX communication can be secured via the standardized **KNX Data Secure** and **KNX IP Secure** mechanisms implemented in newer KNX devices. The security algorithms used in KNX Secure devices are based on security algorithms standardized according to ISO 18033-3 and use recognized encryption according to AES 128 CCM. The following security methods are used to protect the data:

1. **Authentication:** Telegrams are authenticated in such a way that receivers can recognize these telegrams as originating from a trusted sender.
2. **Encryption:** The useful data in the telegram is also encrypted. Encryption makes the actual useful data illegible and impossible to manipulate.
3. **Sequence number:** The sequence number inside the telegram prevents a “replay attack”, i.e., the recording of telegrams and injecting the same telegrams into the system at a later point in time.

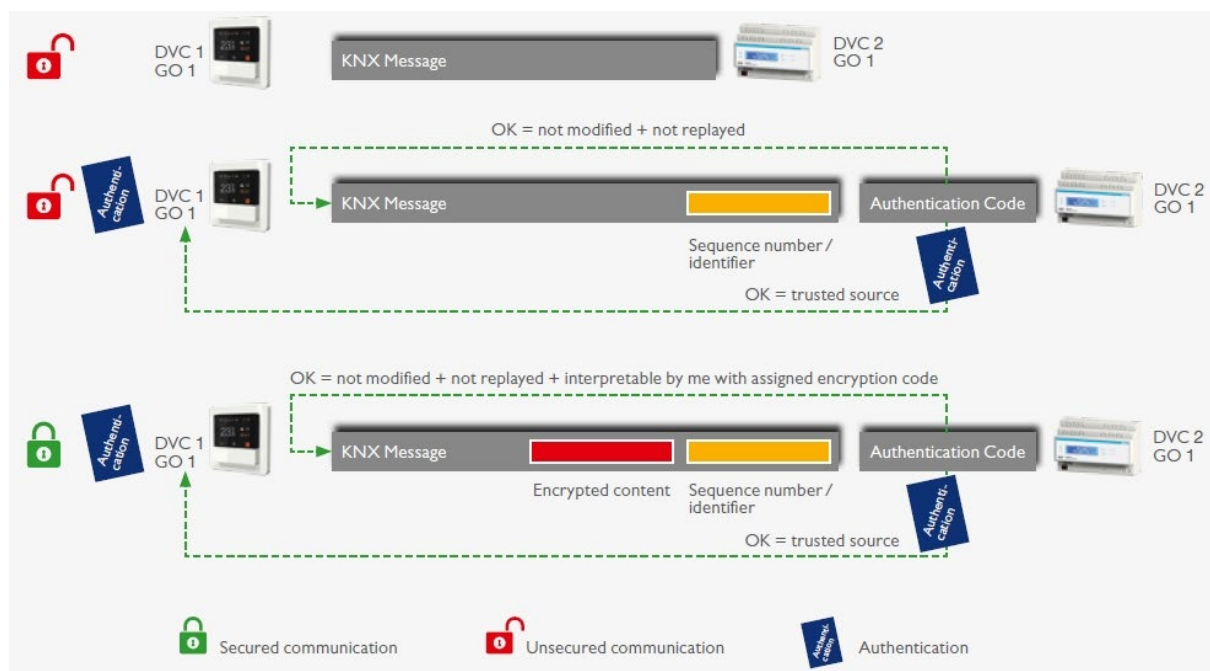


Figure 12: Overview of the KNX Data Secure mechanism

6.1.3.1 Two types of KNX Secure

Two types of KNX Secure currently exist that can be combined with one another and operated in parallel. KNX Secure devices can easily be recognized by a blue colored “shield” in ETS and by the letter “X” printed on the device label.

6.1.3.1.1 KNX Data Secure

KNX Data Secure signs and encrypts communication in the KNX network and ensures secure data transmission of telegrams. The communication during commissioning processes with the ETS as well as the runtime communication between devices and visualizations is secure (provided that all participating bus devices support KNX Data Secure - KNX devices that do not support Data Secure, discard KNX Data Secure telegrams). The concept ensures that all or only selected KNX telegrams are authenticated and encrypted regardless of the medium (TP, RF or IP). In this way communication between sender and receiver can neither be interpreted nor manipulated. Consequently, KNX Data Secure effectively protects the useful data against unauthorized access and manipulation.

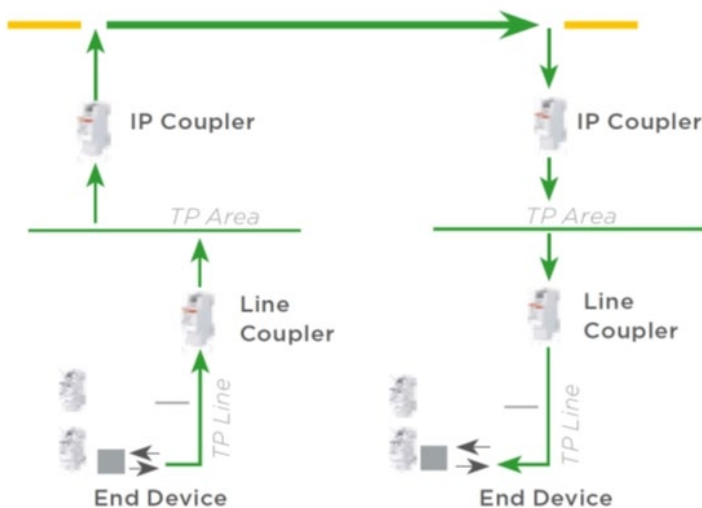


Figure 13: Data Secure (green = secured communication, orange = plain (non-secure) communication)

6.1.3.1.2 KNX IP Secure

KNX IP Secure is used to protect KNX IP Communication. KNX IP Secure encrypts and authenticates telegrams between KNX IP routers on the IP network. In KNX IP Secure the complete frame (and not only the useful data) is encrypted and it can of course only be used on the medium IP.

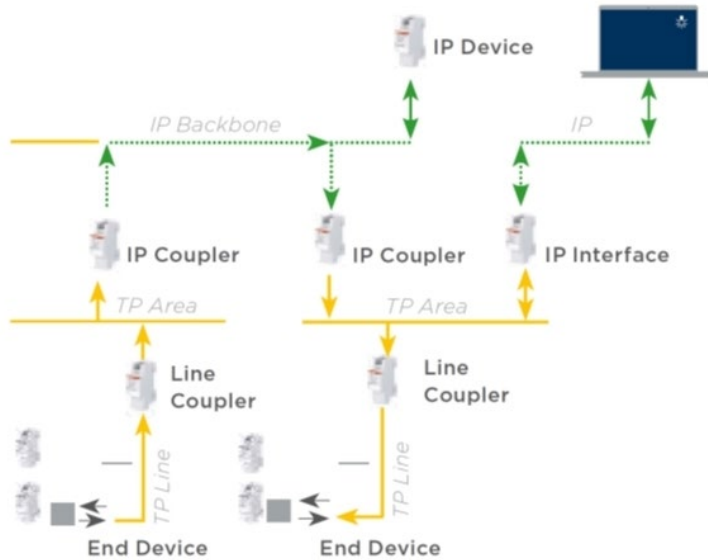


Figure 14: KNX IP Secure (green = secured communication, orange = plain (non-secure) communication)

6.1.3.2 In practice: Secure commissioning in ETS

KNX Data and IP Secure are supported in ETS5.5 and later. In the Properties container of ETS, it is possible to activate or deactivate the “Secure commissioning” property for each KNX Secure device.

In the below-mentioned paragraphs, you will find some important information when configuring KNX Secure devices for the first time.

6.1.3.3 FDSK Key

Each Secure device is shipped with a **Device certificate** consisting of a **unique** unchangeable Factory Device Set up Key (**FDSK**) and a KNX serial number. This certificate is available in the form of a QR code or as a plain code (stuck on to the device or provided separately in the product box). If available as QR code, the ETS user must scan it with the webcam on his PC. If available as plain code, the user must enter this key in ETS during the project design stage or at the latest during the commissioning stage when requested by ETS. Please note that the FDSK is never communicated over the bus.

6.1.3.4 Tool key

Once the FDSK key is entered by the ETS user, ETS will create a device specific **Tool key**. ETS will then send it via the bus to the to be configured device, however by encrypting and authenticating this message with the previously entered FDSK. The device to be configured from then onwards will only accept this Tool key for further configuration with ETS. The FDSK key is no longer used during subsequent communication, unless the device is reset to factory state, after which all the secure data in the device will be erased. This means that the device certificate must be properly archived (by removing the certificate from the device if required).

If a hacker gets physical access to the device, then he may perform a Local Master Reset on the device, which will make the FDSK become active again. This allows him to take control over the device. If the ETS user would further reprogram the device, this will again start with using the FDSK, which can then be sniffed by the hacker revealing the Tool key and all runtime keys.

All Tool keys are stored in the ETS project and are visible to the ETS user in the menu “Project security” of the Reports panel in ETS.

Project Security MySmart House

This report includes security relevant information. Please keep it protected.

Backbone			
Key			
EF1A564CA0CA44E20962E1FE681915A6			
Devices			
Address	Name	Tool Key	Authentication Code
0.0.1	Secure Switching Actuator (D25)	128DD06993594AAD1A393E2D3E6D33A2	d <"_7UK
1.0.0	IP/TP Coupler (D19)	1F9B44947AE53863E746A946566D03C3	+zz*z8R"
1.1.0	TP/TP Coupler (D5)	7359145BD2FF7AF89F8C4F3DA912A5E9	
1.1.1	TP/RF Coupler (D18)	0DAC4222225C86980F827CD40EDD9A0	
1.1.2	Secure KliX (D1)	5535B01CB09261137EA08E9D16028DE8	
1.1.3	Secure Switching Actuator (D3)	1D779C3F60971ACEFE5075D3A77D4D7D	
1.1.16	Secure KliX (D26)	B12D991D0A661FAE83CE1212C91343AF	

Figure 15: Tool Key in menu "Project Security" of Reports panel in ETS

6.1.3.5 Runtime keys

ETS will secretly generate Runtime keys² for group objects that have been assigned to a secure group address. Please note that all communication partners for this group address must be of the type “Secure” (if one group object is not of the type “Secure”, the communication will become plain. These runtime keys are also stored in the ETS project and are visible to the ETS user in the menu “Project security” of the Reports panel in ETS.

6.1.3.6 Project password

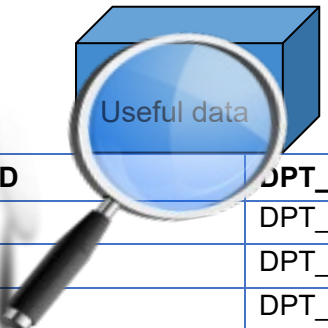
A **project password** is mandatory if there is at least one KNX Secure device in the ETS project. If the project is exported and opened by someone else, the ETS project cannot be opened without this password. This serves to protect the security information (keys) stored in the ETS project.

More in-depth information on the KNX Secure can be found in the chapter “KNX Secure” of the KNX Advanced course.

- ² Also referred to as Group key

7 Standardized datapoint types

7.1 Introduction to standardized datapoint types [E]



DPT_ID	DPT_Name
1.001	DPT_Switch
1.007	DPT_Step
1.008	DPT_UpDown
3.007	DPT_Control_Dimming
5.001	DPT_Scaling
9.001	DPT_Value_Temp
9.004	DPT_Value_Lux
....	

Figure 16: Standardized datapoint types (selection)

To ensure that KNX devices from different manufacturers work together, many datapoint types (DPTs) have been standardized³. These define how sensors and actuators exchange data for functions like dimming lights, setting a clock, or controlling blinds.

The data format and the structure of group objects for both sensors and actuators are standardized in KNX datapoint types.

The combination of several standardized datapoint types is called a functional block. Manufacturers can freely name their group objects, which means:

- DPT_Step might be labeled as "short operation" or "blind operation", depending on the manufacturer.
- The function remains the same, even if the name varies.

The use of the DPT is not limited to one specific application domain. For example, DPT "Scaling" (Type 5.001) can be used for both setting the brightness of a dimmer and adjusting a heating valve position.

The next sections provide examples of various KNX datapoint types. A detailed list of all approved KNX datapoint types can be found in the KNX Help Center:

<https://support.knx.org/hc/en-us/articles/15392604906514-Interworking-Datapoint-types>

³ Nowadays, manufacturers also contribute by assigning DPTs to group objects in their modern application programs, ensuring better compatibility and seamless integration with other KNX devices.

7.2 On/Off (1.001) [E]



DPT_Switch (1.001)		
Size	Encoding	Meaning
1 Bit	0	• Off
	1	• On

DPT_Bool (1.002)		
Size	Encoding	Meaning
1 Bit	0	• False
	1	• True

DPT_Enable (1.003)		
Size	Encoding	Meaning
1 Bit	0	• Disable
	1	• Enable

Figure 17: DPT_Switch (1.001), ...

DPT_Switch (1.001) is a 1-bit datapoint type used for turning actuators on or off. Other 1-bit datapoint types exist for different functions, such as:

- Boolean operations (1.002): For logical operations.
- Enable/Disable (1.003): For enabling/disabling a function.

Features like inversion, time delay, and toggle switching are not part of the datapoint type itself. These are parameters of the functional block specification where this DPT is used (e.g., Functional Block Light Switch).

A switching function can also include a status or feedback object to indicate the current state (ON/OFF) of the actuator. To enable status feedback, you need to create a group address that links the status object of the rocker switch with the feedback object of the actuator relay. Both the status and feedback objects use 1-bit data length.

7.3 Functional Block “Sunblind Actuator Basic”



DPT_UpDown (1.008)		
Size	Encoding	Meaning
1 bit	0	• Up
	1	• Down

DPT_Step (1.007)		
Size	Encoding	Meaning
1 bit	0	• Decrease, Stop
	1	• Increase, Stop

Figure 18: Functional block "Sunblind Actuator Basic"

This functional block is designed for controlling solar protection devices such as blind and shutter drives. It requires at least two group objects with the following datapoint types:

- **Up/Down (DPT 1.008) – Long press on a rocker:**
 - Starts moving the shutter/blind from a stopped state.
 - If already moving, it possibly changes the direction after a turnaround break.
- **Step/Stop (DPT 1.007) – Short press on a rocker:**
 - If moving: Stops the motion.
 - If stopped: Adjusts blind slats by one step per press.

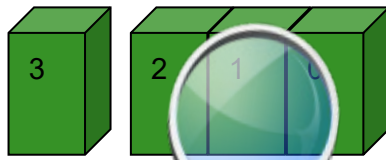
The above group objects should **NOT** respond to Read telegrams on the bus because if a device responds to a read request, it may unintentionally:

- Stop moving drives.
- Start moving again if the drive is currently stopped.

The solution is to **disable the Read flag** in ETS for the relevant group objects (both in sensors and actuators). This is especially important for central functions that control multiple blinds/shutters at once.

Note: Although DPT_Switch and DPT_Step or DPT_UpDown (or any 1-bit DPT) use the same 1-bit information (and could in fact be linked in ETS), they should only be linked if the linking makes functional sense e.g., Lighting off, move up sun shading (DPT_Switch and DPT_UpDown).

7.4 Functional block “Dimming” [E]



DPT_Control_Dimming (3.007)		
Size	Encoding	Meaning
4 Bit	Bit 3: 0	• Decrease (darker)
	Bit 3: 1	• Increase (brighter)
	Bit 0 – 2	• Number of dimming steps



0 – 255 Dimming values

DPT_Scaling (5.001)		
Size	Range	Meaning
1 byte	0 – 100%	Dimming values

Figure 19: Functional block “Dimming”

The Functional Block “Dimming” in KNX consists of three key datapoint types for controlling the brightness:

- **DPT_Switch (1.001): 1-bit object (On/Off Switching)**
 - Turns the light on or off without adjusting brightness.
 - Example: A standard light switch function (See §6.2).
- **DPT_Control_Dimming (3.007): 4-bit object (Relative Dimming)**
 - This allows gradual dimming of the actuator relative to the current brightness level.
 - The dimming direction (brighter or darker) is determined by bit 3 (sign bit) of the useful data.
 - Bits 0 to 2 define the dimming step size:
 - The smallest step is 1/64th of 100% brightness (1% per step in ETS group monitor).
 - Example: Holding a button dims up/down until released.
- **DPT_Scaling (5.001) - 8-bit object (Absolute Dimming)**
 - Sets brightness to a specific value between 0% (minimum) and 100% (maximum).
 - The brightness level is set with a resolution of approximately 0.4% (since there are 256 possible values from 0 to 255).
 - Depending on the manufacturer's implementation, the same function might also allow:
 - Turning the light ON if the value is between 0.4% and 100%.
 - Turning the light OFF if the value is 0%.

- Example: A slider control on a touch panel sets brightness to 50%.

Group object support in devices for dimming:

- Dimming sensor
 - Must support at least:
 - DPT_Switch (1.001): Turns lights on/off.
 - DPT_Control_Dimming (3.007): Sends relative dimming commands.
- Dimming actuator
 - Must support:
 - DPT_Switch (1.001): Receives on/off commands.
 - DPT_Control_Dimming (3.007): Receives relative dimming commands.
 - **DPT_Scaling (5.001)**: Receives absolute brightness values.

7.4.1 Difference between relative dimming and absolute dimming

Feature	Absolute dimming (DPT_Scaling [5.001])	Relative dimming (DPT_Control_Dimming [3.007])
Function	Sets a fixed brightness level.	Increases or decreases brightness from the current level.
Data size	1 byte (8-bit, 256 values).	4-bit (only dim up/down commands)
Range	0% to 100%.	Changes the brightness relative to current level.
Example use case	Setting brightness to 50%.	Holding a button to gradually dim up/down.

7.5 KNX data format for Floating Point Values (9.0xx)

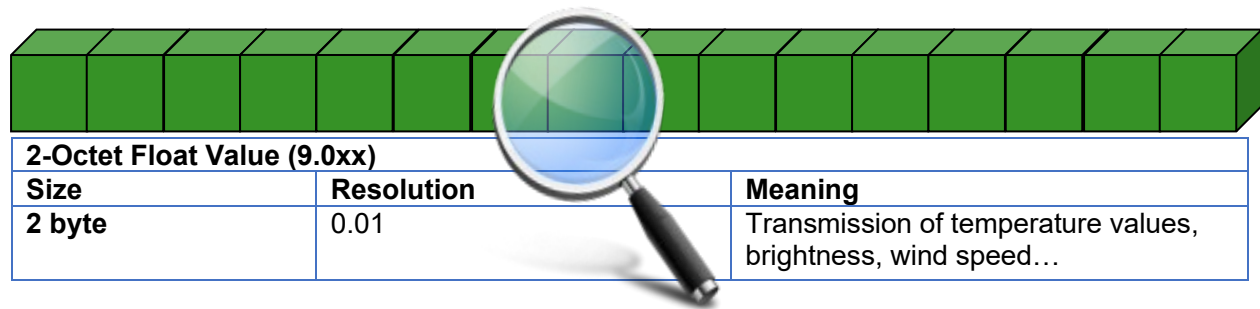


Figure 20: 2-octet float value (9.0xx)

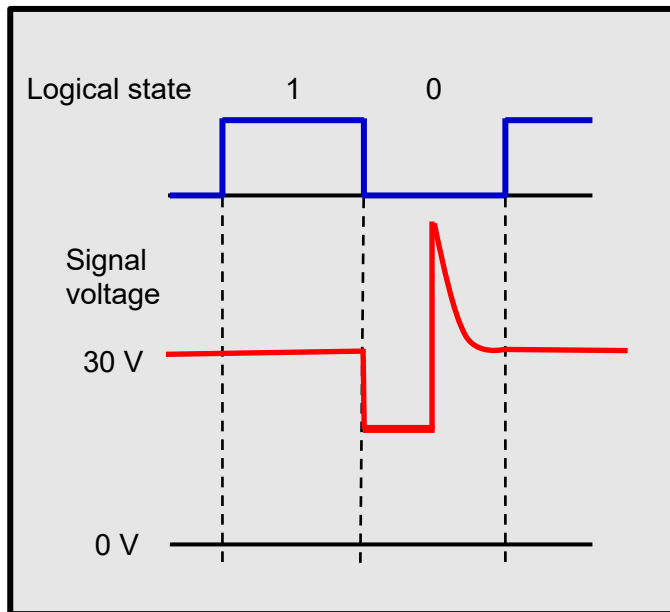
This data format is used to transmit positive or negative floating-point values with a maximum resolution of 0.01. It allows for precise measurements and is commonly used in various KNX datapoint types such as:

- **DPT_Value_Temp (9.001):** Used for measuring and transmitting room temperature (°C).
- **DPT_Value_Wsp (9.005):** Used for measuring and transmitting wind speed (m/s).

The group object size is 2 bytes (16 bits).

8 KNX TP bit structure

0 and 1 are the two logical states a bit can have.



A Bit in KNX TP:

Logical 1 → no change of bus voltage
Logical 0 → bus voltage shortly varied



This implies that if multiple devices transmit at the same time, the device with logical state 0 will prevail and may continue sending.

Figure 21: TP bit structure

In KNX TP communication, each bit has two possible logical states: 0 and 1. The system follows specific rules to define these states based on signal voltage and current flow:

- Logical state "1": No signal voltage, no change of bus voltage.
- Logical state "0": Signal voltage present, bus voltage shortly varied.

This means that if multiple devices transmit at the same time, the device with logical state "0" will always prevail and may continue sending.

9 Bus access and telegram collision in KNX TP

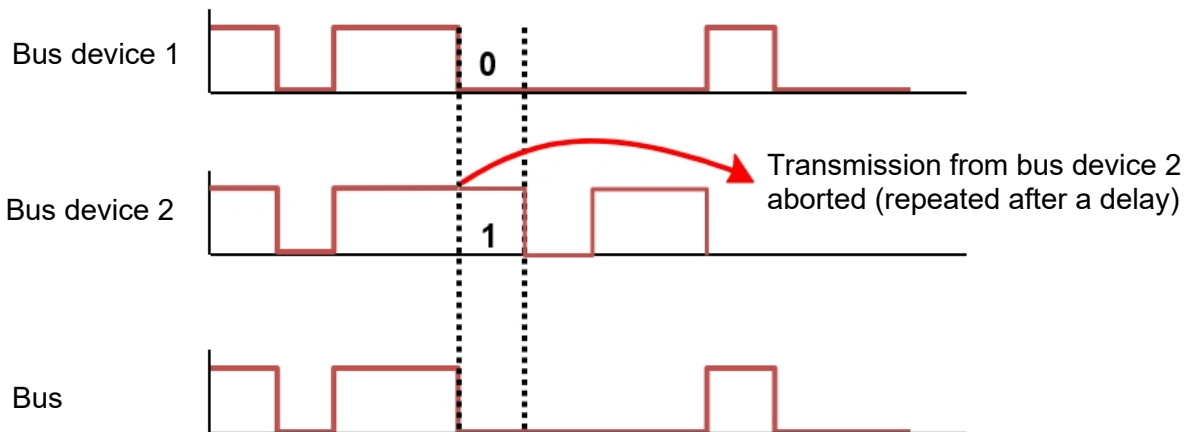


Figure 22: Telegram collision

In KNX TP, bus access is not controlled by a master/slave system. Instead, devices can transmit whenever the bus is free. This communication is random, without fixed time slots.

How KNX TP handles multiple devices sending at the same time (CSMA/CA):

- KNX TP uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) to prevent transmission conflicts.
- Devices listen to the bus before sending. If the bus is free, they start transmission.
- While transmitting, devices also monitor the bus:
 - If a device detects another transmission with a logical "0" (voltage varied) while having a logical "1" (voltage unchanged), it stops transmitting to avoid conflict.
 - The device that stopped the transmission, waits for the current transmission to finish, listens to the bus, and then tries again to send its telegram.

The CSMA/CA system thus ensures that the device sending a logical "0" (dominant state) wins, while the others pause and retry after the transmission is complete. This process ensures smooth communication without slowing down data flow.

When a device sends a telegram with higher priority, all devices with lower-priority telegrams must wait until the high-priority telegram has been fully transmitted. After the transmission is complete, the waiting devices will retry sending their telegrams.

10 Symmetrical signal transmission in KNX TP

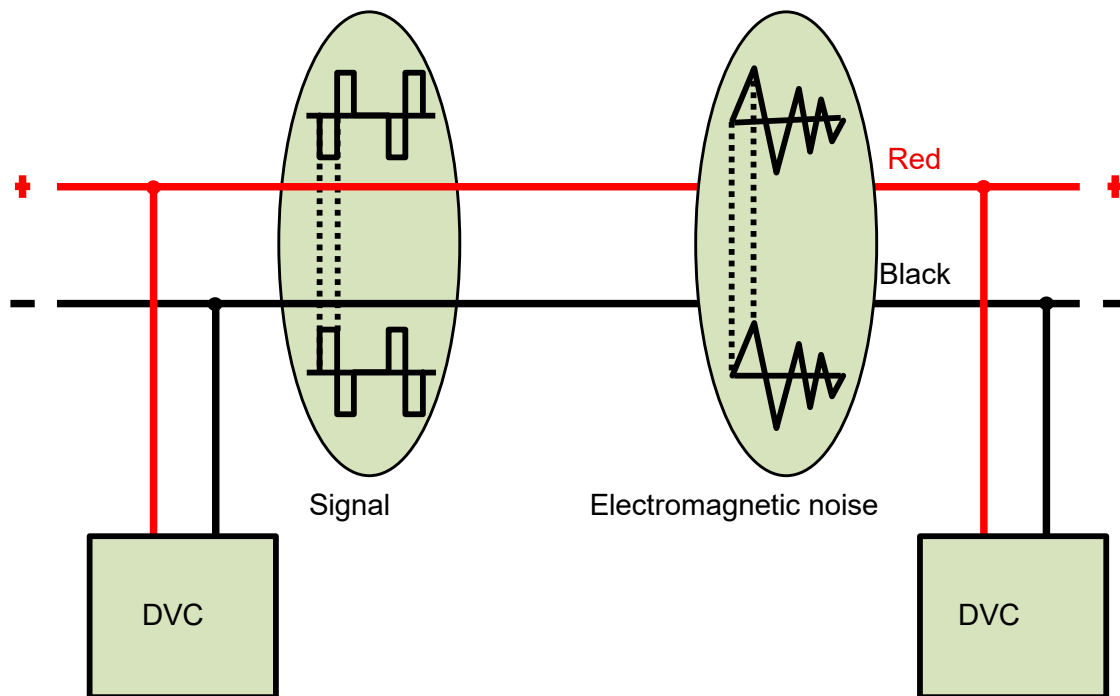


Figure 23: Symmetrical signal transmission

In KNX TP, data is transmitted symmetrically over a pair of twisted wires. Both wires carry the signal, but none are connected to ground (PE) or a fixed potential. Symmetrical transmission is used to:

- Reduce noise interference:
 - External electromagnetic noise affects both wires equally.
 - Only the voltage difference between both wires is evaluated. This will make the signal immune to external noise.
- Ensure reliable communication in electrically noisy environments.

11 Superimposing data and supply voltage in KNX TP

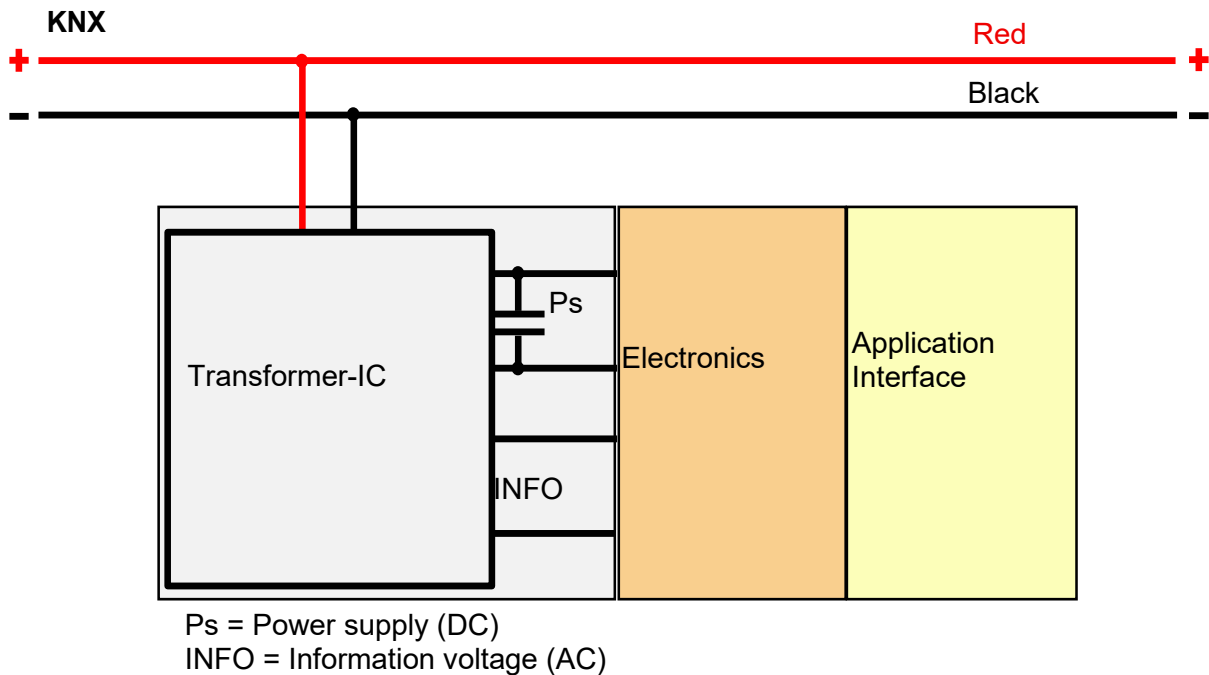


Figure 24: The transformer-IC in the bus device separates DC supply voltage and AC information voltage

In KNX TP, both data and power are transmitted over the same pair of wires. Data is transmitted as AC voltage, while power is supplied as DC voltage. The data signal (AC) is superimposed onto the supply voltage (DC). A transformer-IC inside each bus device separates these two components:

- Extracts DC voltage for power.
- Reads AC voltage for communication.

12 Connection of the power supply unit to the TP bus

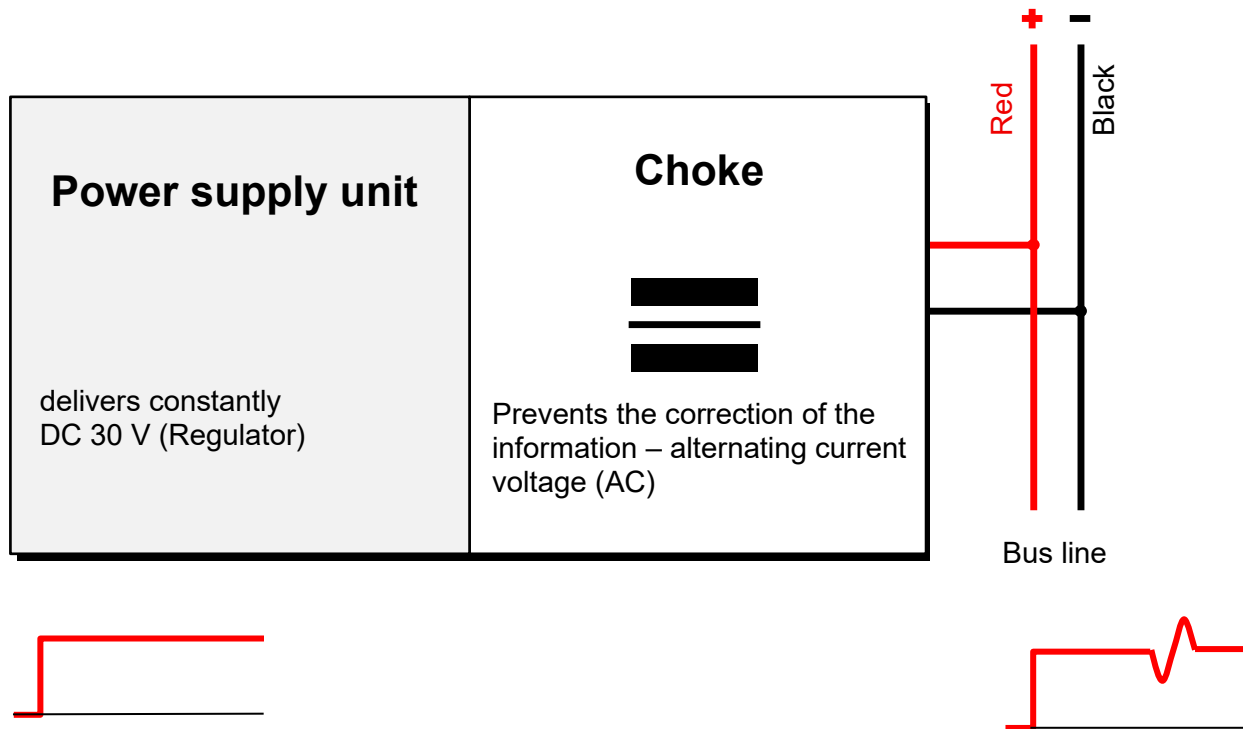


Figure 25: Connection of power supply to TP bus

In a KNX TP system, the power supply unit (PSU) feeds the bus via a choke. This choke plays a critical role in ensuring stable operation:

Why is a choke needed?

- The power supply includes a voltage regulator to maintain a stable DC 30 V.
- If the bus were connected directly to the power supply, the voltage regulator would also try to correct the AC data signals, leading to a conflict ("tug of war") between the sending bus device and the regulator.
- The choke (inductor) slows down the regulator's response, allowing temporary voltage deviations needed for data transmission.

Role of the choke in signal generation:

- The sending bus device generates only the first (negative) half of the AC signal pulse.
- The choke generates the second (positive) half, ensuring that the final AC signal has no DC offset.
- This symmetrical AC signal is essential for proper data decoding by receiving devices.

13 Cable length limits in KNX TP

13.1 Introduction [E]

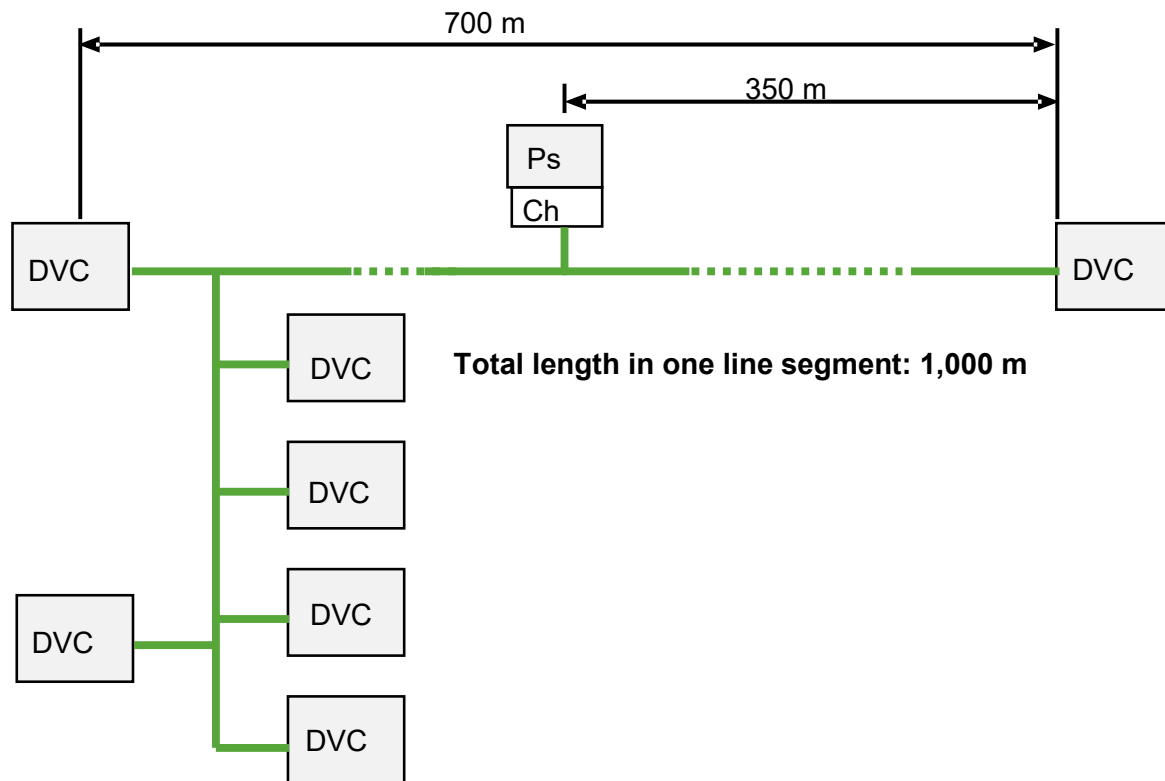


Figure 26: cable lengths

To ensure reliable communication, KNX TP installations must follow specific cable length limits. Maximum cable lengths in a KNX TP line segment:

Power supply unit – Bus device	Max. 350 m
Bus device – Bus device	Max. 700 m
Total bus line segment cable length	Max. 1,000 m
Distance between two power supplies in one line segment	Refer to manufacturer's instructions

Why cable length limits matter:

- Exceeding the maximum length can cause data transmission issues that are difficult to diagnose.
- Staying within these limits ensures stable communication and power distribution across the KNX TP network.

13.2 Cable length between TP power supply unit – TP bus device

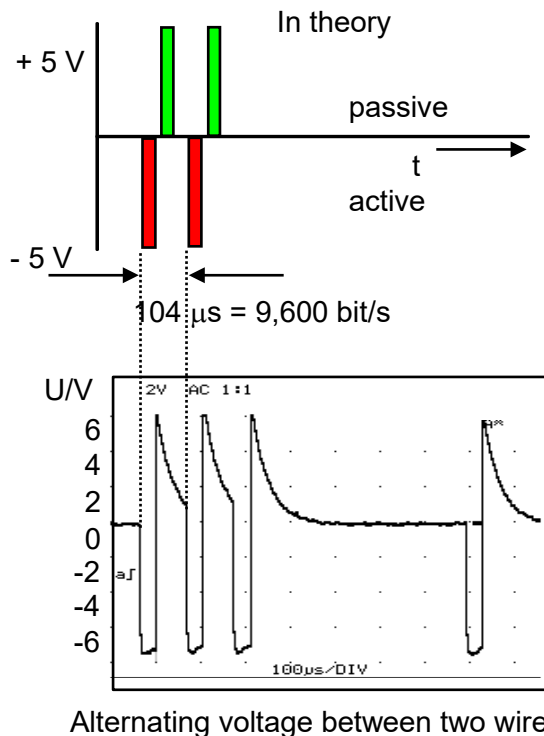


Figure 27: Cable length between TP power supply unit - TP bus device

A bus device only generates the first (negative) half of the signal pulse. The choke in the power supply unit generates the second (positive) half (called the equalization pulse). Because the choke is essential for properly forming the signal, the distance between a bus device and the power supply must not exceed 350 meters.

If a bus device is too far from the power supply, the equalization pulse may weaken, causing signal instability or loss. Ensuring devices are within 350 meters maintains proper communication and reliable operation.

13.3 Cable lengths between two TP bus devices

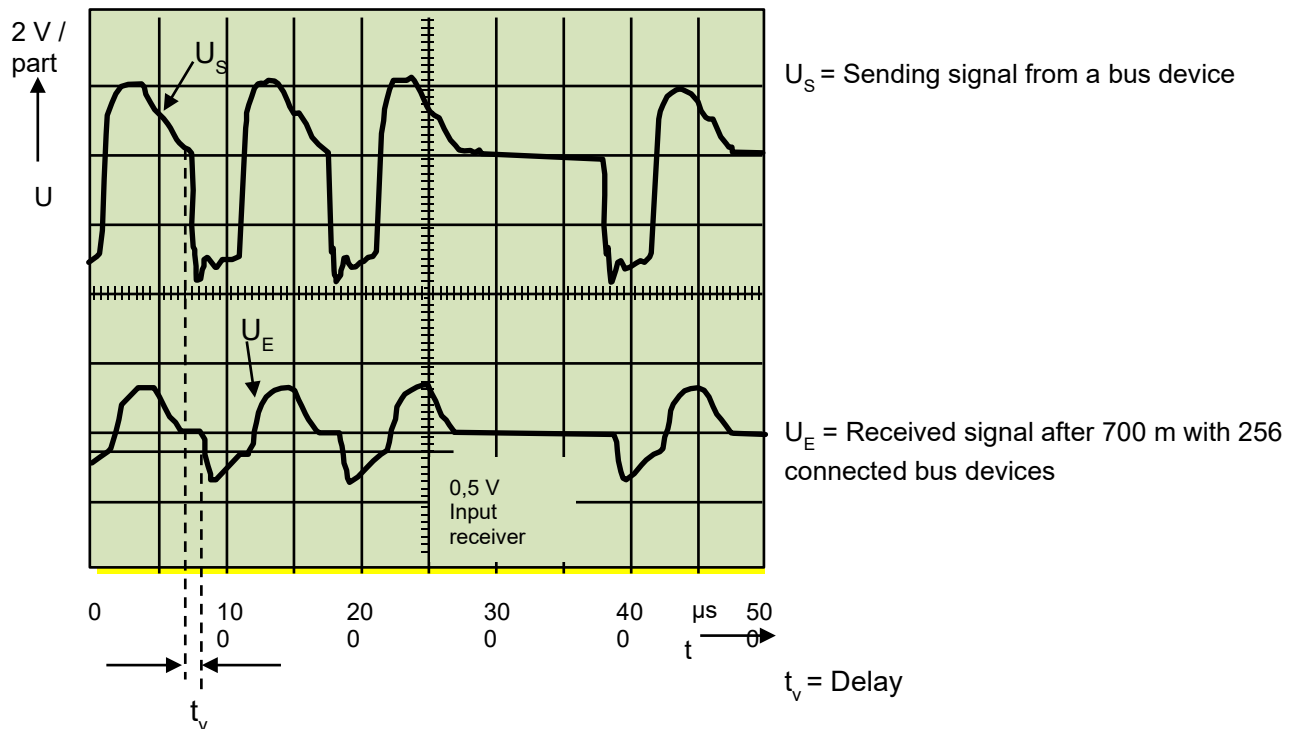


Figure 28: Cable lengths between two TP bus devices

In a KNX TP system, signals take a certain amount of time to travel through the cable. If two bus devices transmit at the same time, the system detects collisions and resolves them up to a distance of 700 m because in this case the signal delay time (t_v) remains below 10 μs . If devices are too far apart (> 700 m), the signal delay could exceed 10 μs , making it difficult to detect and correct collisions in time.

13.4 Total cable length per TP line segment

In a KNX TP system, the signal quality can be affected by the cable length and the number of connected bus devices. The longer the cable, the more the signal weakens due to:

- **Cable capacity:** The continuous charging and discharging of the cable affects signal transmission.
- **Signal rounding:** The edges of the signal become less sharp, making it harder to read.
- **Resistive load:** The bus cable and the connected bus devices have a certain ohmic resistance, which lowers the signal strength.

To maintain reliable data transmission, the total cable length per line segment must not exceed 1,000 meters.

The maximum number of bus devices in a line segment depends on their power consumption (see chapter Topology for more information).